

Traditional herbal medicine-derived sulforaphene promotes mitophagic cell death in lymphoma cells through CRM1-mediated p62/SQSTM1 accumulation and AMPK activation

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ABSTRACT

Sulforaphene (LFS-01) is the major chemical constituent of *Raphanus sativus*, a medicinal herb used for over a thousand years in traditional Chinese medicine. Here we identified that LFS-01 can selectively eradicate lymphoma cells while sparing normal lymphocytes by triggering concomitant mitophagy and apoptosis. We demonstrated that LFS-01 can retain Nrf2 in the nucleus by covalently modulating CRM1 and consequently up-regulate p62/SQSTM1, an essential structural component of the autophagosomes during mitophagic process. We found that LFS-01 treatment also stimulated AMPK and thereby inhibited the mTOR pathway. On the contrary, we revealed that AMPK inhibition can severely impair the LFS-01-mediated mitophagy. Transcriptomic studies confirmed that 15 autophagy-associated genes such as *p62/SQSTM1*, *VCP* and *BCL2* were differentially expressed after LFS-01 treatment. Furthermore, protein interactome network analysis revealed that the events of apoptosis and the assembly of autophagy vacuole were significant upon LFS-01 exposure. Lastly, we found that LFS-01 exhibited strong efficacy in xenograft mouse model yet with the lack of apparent toxicity to animals. We concluded that LFS-01 triggered mitophagic cell death via CRM1-mediated p62 overexpression and AMPK activation. Our findings provide new insights into the mechanism of action for LFS-01 and highlight its potential applications in treating major human diseases.

1. Introduction

The medicinal effects of *Lai Fu Zi* (*Raphanus sativus*) have been documented since the Song dynasty for over one thousand years in traditional Chinese medicine to treat symptoms such as coughing and food stagnation [1]. The major active component of *Lai Fu Zi* is a naturally-derived isothiocyanate compound named as sulforaphene (LFS-01, Fig. 1a), which has recently raised considerable research interests worldwide for its excellent therapeutic activities yet with minimal side effects. However, the exact mechanism of action for LFS-01 in malignant cells is still poorly understood and remains to be clearly defined. Interestingly, we found that LFS-01 can induce apoptosis in numerous lymphoma cell lines while sparing normal lymphocytes

isolated from healthy donors. More interestingly, the effects of LFS-01 in lymphoma cells are only partially suppressed by caspase inhibitors, strongly implicating that LFS-01 may also exert its potent antitumor effects through caspase-independent cellular death.

Mitophagy refers to the selective degradation of defective mitochondria by autophagy in response to stress conditions [2]. This cargo-specific autophagy involves a few distinct steps to selectively engulf damaged mitochondria into double-membraned vesicles called autophagosomes, in which LC3-II and p62 act as essential structural components [3]. Noteworthy, a very recent study particularly suggests that p62 is crucial for mitophagic clearance and depletion of p62 in cells directly disrupts mitophagy [4]. Furthermore, mammalian target of rapamycin (mTOR) kinase was recently identified as a master

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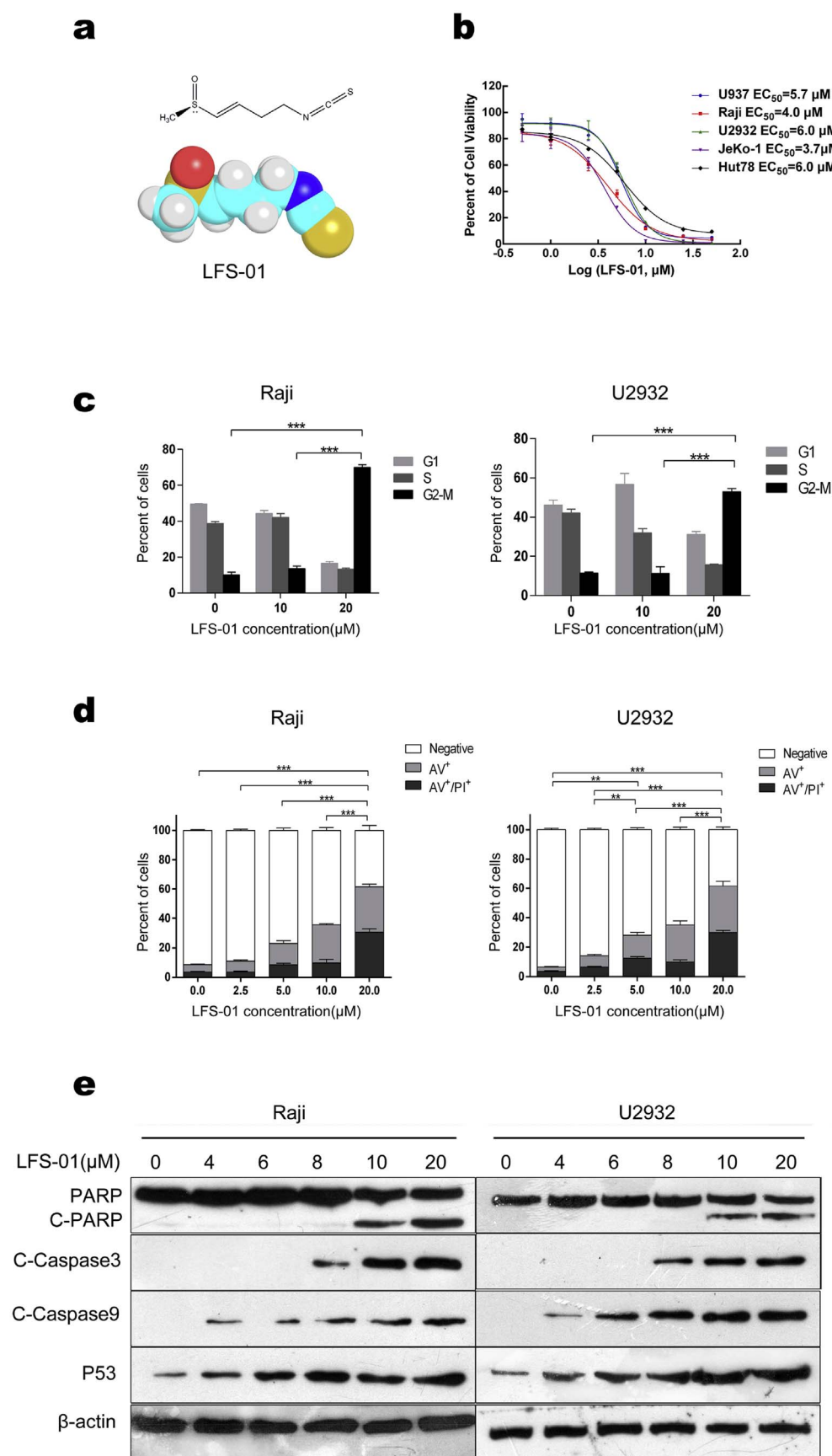


Fig. 1. LFS-01 inhibits lymphoma cell growth by inducing cell cycle arrest and apoptosis. (a) Chemical structure and CPK model of LFS-01. (b) Suppression of lymphoma cells growth by LFS-01. Different types of lymphoma cells were treated with the indicated concentrations of LFS-01 for 72 h. Cell viability was measured by CCK8 kit in triplicate wells. (c) LFS-01 induces cell cycle arrest in G2-M phase. Raji and U2932 cells were treated with the indicated concentrations of LFS-01 for 24 h. Cells were then harvested, stained with propidium iodide (PI) and analyzed by flow cytometry. * $P < .05$, ** $P < .01$, *** $P < .001$ by one-way ANOVA, followed by Dunnett's test or Tukey's test. $N = 3$. Error bars = SEM. (d) LFS-01 induces apoptosis. Raji and U2932 cells were treated with the indicated concentrations of LFS-01 for 24 h. Cells were then harvested, stained with Annexin V and PI and analyzed by flow cytometry. * $P < .05$, ** $P < .01$, *** $P < .001$ by one-way ANOVA, followed by Dunnett's test or Tukey's test. $N = 3$. Error bars = SEM. AV, Annexin V. PI, Propidium iodide. (e) Representative Western blot showing levels of P53 and apoptosis proteins. Cells were treated with LFS-01 at the indicated concentrations for 24 h. Whole cell proteins were extracted and subjected to immunoblotting analysis. Actin serves as loading control.

regulator of autophagy or mitophagy [5]. Once activated, mTOR can inhibit autophagy or mitophagy followed by reformation of lysosomes [6]. Conversely, AMP-activated protein kinase (AMPK), a key energy sensor, has been recently found to promote mitophagy via negative regulation of mTOR in response to nutrient deprivation or environmental stress [7,8]. Accumulating evidences suggest that mitophagy plays an important role in the regulation of malignant cell survival and metabolism under low-energy or stress conditions. Interestingly, mitophagy could be pro- or anti-tumorigenic depending on tumor stage. For instance, decreased mitophagy may enable dysfunctional mitochondria to persist and thereby generate increased tumor-promoting reactive oxygen species (ROS) or other tumorigenic mitochondrial signals [9]. Consequently, agents that induce mitophagy could exert anti-tumor effects through the clearance of dysfunctional mitochondria. In addition, it has been shown that mitophagy-deficient mice are susceptible to increased tumorigenesis [10]. Such initial findings have raised the possibility that mitophagy could be a novel target for the anti-neoplastic responses of cancer therapeutics.

In this study, for the first time, we demonstrated that herb-derived LFS-01 induces concomitant mitophagy and apoptosis in lymphoma cells. Through various biochemical studies, we showed that LFS-01 directly targets the sulfhydryl (-SH) group of a conserved cysteine in the nuclear export signal (NES)-binding groove of CRM1 protein and leads to the nuclear retention of Nrf2, which consequently upregulates p62, an essential structural component of the autophagosomes towards mitophagic clearance. Our RNA-seq experiments also confirmed that *p62/SQSTM1* was significantly overexpressed upon LFS-01 exposure. Furthermore, we showed that LFS-01 treatment activates AMPK/mTOR pathway and leads to mitophagic signaling in lymphoma cells. We revealed that LFS-01 administration can significantly suppress tumor growth in a xenograft model with minimal toxicity. Our findings establish a critical role for mitophagy in mediating the biological effects of LFS-01 and suggest that LFS-01 could be used as a prototype for the development of next-generation cancer therapeutics targeting on mitophagy.

2. Materials and methods

2.1. Cell lines, antibodies and reagents

RPMI 1640 medium, DMEM medium and FBS (fetal bovine serum) were purchased from Hyclone (Thermo Scientific). Human lymphoma cell lines U937, HUT78, Raji, JeKo-1 and U2932 were purchased from ATCC. U937, Raji and U2932 cell lines were maintained in RPMI 1640 medium supplemented with 10% FBS, JeKo-1 in RPMI 1640 medium, and HUT78 in DMEM medium with 20% FBS. Antibodies against RanBP1 (sc-28576) and CRM1 (sc-5595) were obtained from Santa Cruz Biotechnology, Texas, USA. Other antibodies were purchased from Cell Signaling Technology: anti- β -actin (#3700), anti-P53 (#2524), anti-PARP (#9532), anti-cleaved caspase 3 (#9664), anti-cleaved caspase 9 (#7237), anti-LC3A (#4599), anti-HA-Tag (#3724), anti-mTOR (#2983), anti-p-mTOR (#2971), anti-AMPK (#5832), anti-p-AMPK (#2535), anti-p62 (#8025), anti-Keap1 (#8047), anti-Nrf2 (#12721). Alexa 488-conjugated goat anti-rabbit antibody and Alexa 594-conjugated goat anti-mouse antibody were obtained from Invitrogen Life Technology (Invitrogen, CA, USA). Sulforaphene (LFS-01, purity $\geq 97\%$) was prepared and purchased from Hangzhou Tianhong Biotechnology.

Other compounds and kits used were RNase A (Coolaber Co Ltd, Beijing, China), Propidium Iodide/PI (Sigma-Aldrich, St. Louis, MO, USA), z-VAD-FMK (Selleckchem, Huston, USA), 3-MA (Selleckchem, Huston, USA), dorsomorphin 2HCl (Selleckchem, Huston, USA) and Hoechst 33258 (Coolaber Co Ltd, Beijing, China). According to manufacturers' reports, the purity of all compounds used in the experiments is $\geq 95\%$.

2.2. PBMCs separation

Normal human peripheral blood mononuclear cells (PBMCs) were obtained from blood samples collected from healthy volunteers. Approval was obtained from The First Affiliated Hospital of Dalian Medical University institutional review board for these studies. Peripheral blood mononuclear cells were isolated by density gradient centrifugation over Histopaque-1077 (Sigma Diagnostics, St. Louis, MO, USA) at 400 g for 30 min. Isolated mononuclear cells were washed and assayed for total number and viability using Trypan blue exclusion. PBMCs were suspended at 8×10^5 /mL and incubated in RPMI 1640 medium containing 10% FBS in 24-well plates.

2.3. Plasmid and cell transfection

The HA-tagged wild-type CRM1 and CRM1 mutant Cys528 to Ser plasmids were kindly provided as gifts by Dr. Chang Chen from the Institute of Biophysics, Chinese Academy of Science. Raji cell populations stably expressing wild-type CRM1 or mutants were transfected with Lipofectamine 2000 (Invitrogen) and enriched by selection by 500 μ g/mL G418 for 3 weeks according to the manufacturer's protocol.

2.4. Cell viability assay

Cell viability assays were performed as previously described [11]. Human lymphoma cell lines U937, HUT78, Raji, JeKo-1 and U2932 were seeded into 96-well plates and treated with LFS-01 in concentrations of 0–50 μ M for 72 h. Cell viability was evaluated using the WST-8-based Cell Counting kit-8 (Beyotime), which was added to the wells and incubated for 3 h. The absorbance of wells at 450 nm (reference wavelength 610 nm) was measured with a microplate reader (Infinite F50, Tecan).

2.5. Cell cycle arrest and cell apoptosis analysis

Raji and U2932 cells were seeded into six-well plates at a density of 2×10^6 cells per well followed by addition of LFS-01 or DMSO for 24 h. After treatment, cells were collected and fixed in cold 70% ethanol overnight. Next day, cells were washed twice with PBS and stained with PI (50 μ g/mL) containing RNase (100 μ g/mL) in a dark place for 30 min. Subsequently, cell cycle arrest was detected on FACS Calibur flow cytometer (Becton-Dickinson) and the results were analyzed with ModFit LT 3.0 software. Cell apoptosis was detected using Annexin V Apoptosis Detection Kit FITC (eBioscience) according to the manufacturer's protocols. Apoptosis results were obtained and analyzed on FACS Vantage flow cytometer (Becton-Dickinson).

2.6. Immunofluorescence

Lymphoma cells were plated on a 24-well plate at a density of 1×10^5 cells per well. Following the indicated treatments, cells were washed with PBS, adhered onto adhesion microscope slide (Citotest; China), then fixed for 20 min with 4% paraformaldehyde in PBS. Next, cell membranes were permeabilized by treatment with 0.3% Triton X-100 in PBS for 20 min. After blocking with 5% bovine serum albumin (BSA) in PBS for 1 h at 37 °C, cells were treated with primary antibodies in blocking buffer for 1 h. Anti-mouse Alexa 594 and anti-rabbit Alexa 488 were used as secondary antibodies. Cell nuclei were stained with 10 μ g/mL Hoechst 33258 for 20 min. After washing, photomicrographic images were recorded with confocal laser scanning microscope Fluoview FV10i (Olympus, Japan).

2.7. Immunohistochemical analysis

For immunohistochemistry, 4 μ m thick sections of tumor tissues were deparaffinized and rehydrated, and antigen retrieval was carried

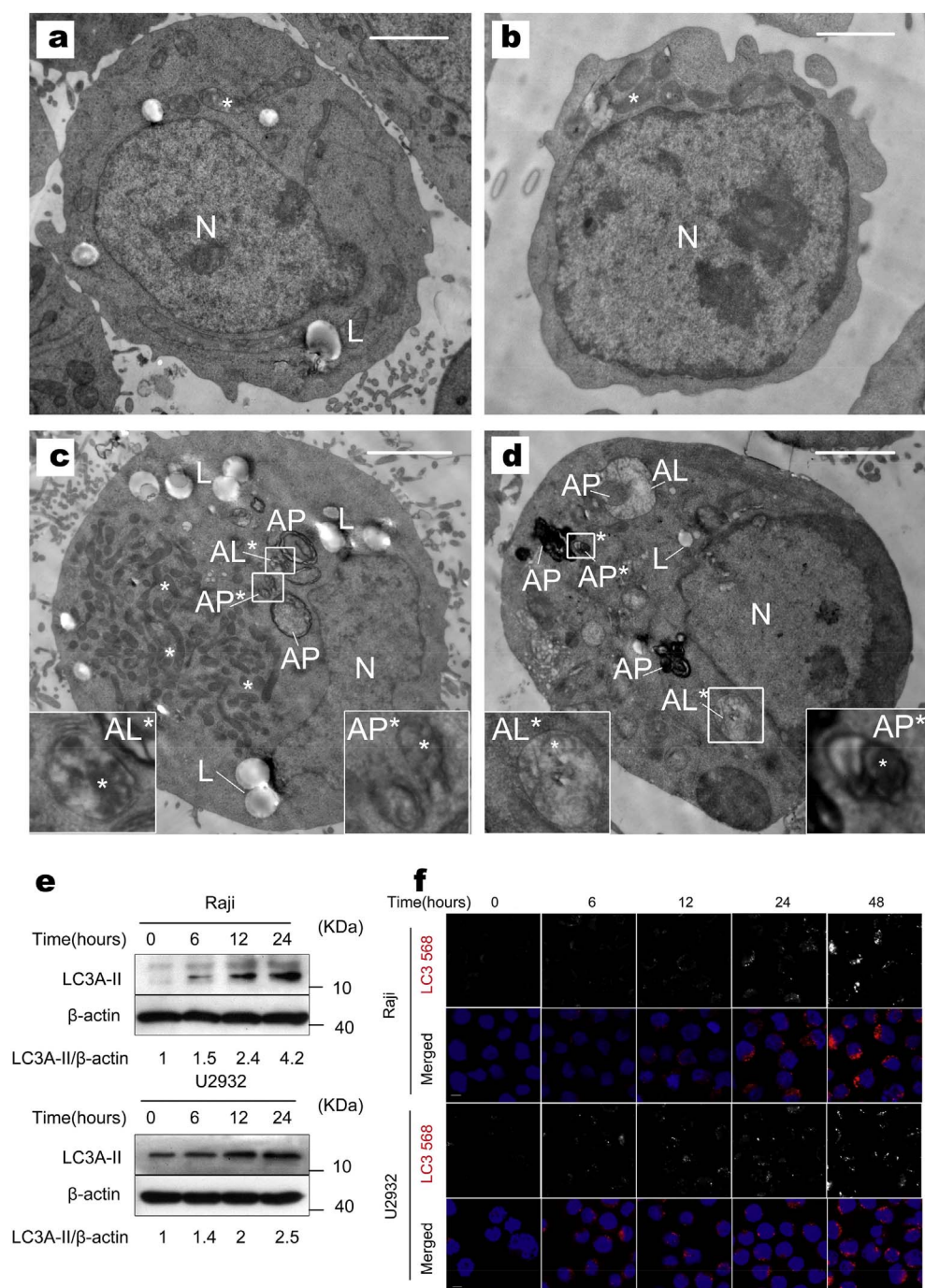
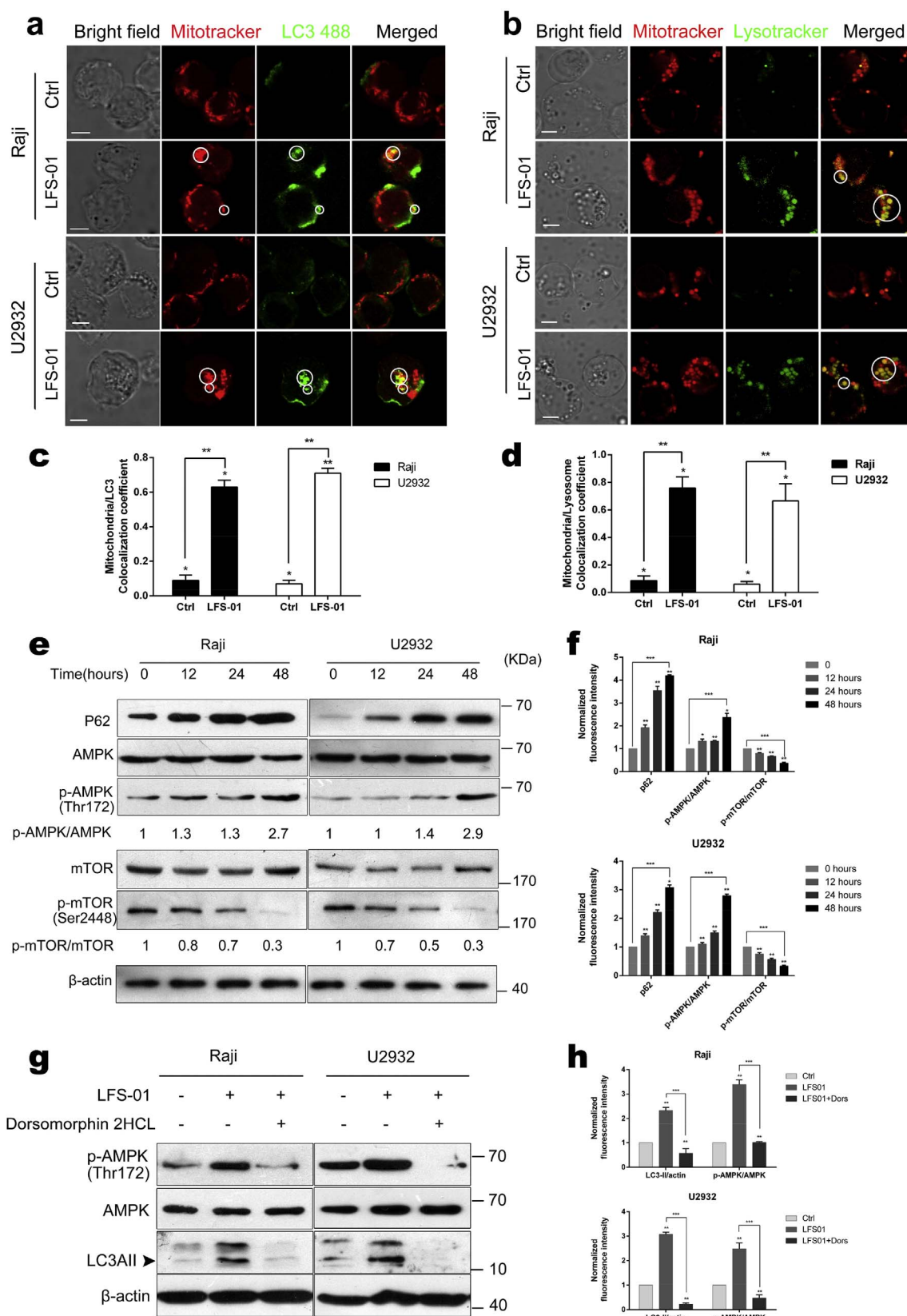


Fig. 2. LFS-01 induces autophagy in Raji and U2932 cells. (a–d) Electron microscopy of mitochondria in Raji and U2932 cells induced by LFS-01. a and b, Raji and U2932 cell treated with DMSO; c and d, Raji and U2932 treated with 10 μ M LFS-01 for 24 h. N, nucleus; L, lipid droplet; asterisks, mitochondria; AP, autophagosomes; AP*, autophagosome with mitochondria; AL, autolysosome; AL*, autolysosome with mitochondria. Scale bar, 2 μ m. (e) Representative Western result showing level of LC3A-II with actin as loading control. The cells were treated with 10 μ M LFS-01 for 0, 6, 12, and 24 h. (f) Confocal microscopy shows time-dependent increase of LC3 (red) punctate staining pattern when treated with 10 μ M LFS-01 for 24 h compared with control, with Hoechst 33258 staining of the nucleus (blue). Scale bar, 5 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

out by microwave treatment of the slides in 50 mM Tris-EDTA (pH 9.0) for 20 min. Endogenous peroxidase was quenched with 3% hydrogen peroxide for 15 min at room temperature. Non-specific binding was blocked with 10% normal goat serum for 15 min. The slides were incubated with rabbit monoanti-LC3A antibody overnight at 4 $^{\circ}$ C. Sections were then incubated with 1–3 drops SignalStain[®] Boost Detection Reagent (HRP, Rabbit) as needed. The incubation was performed in a humidified chamber for 30 min at room temperature. Each section was washed three times for 5 min. Add 1 drop (30 μ l) SignalStain[®] DAB Chromogen Concentrate to 1 mL SignalStain[®] DAB Diluent and mix well before use. Apply 100–400 μ l SignalStain[®] DAB to each section and monitor closely. Immerse slides in dH₂O and slides were counterstained with hematoxylin.

2.8. Western blot

Cells (2×10^6) were grown in 6-well plates and exposed to the designated concentrations of LFS-01 for 72 h or exposed to EC₅₀ concentration of LFS-01 for 12, 24 and 48 h, followed by extraction of the whole cell protein. Western blot was performed according to standard protocol. Protein (40–100 μ g) was fractionated on a 10%–15% acrylamide denaturing gel and transferred onto a PVDF membrane. The membrane was blocked with 5% nonfat dry milk in TBST (50 mM/L Tris (pH 7.5), 150 mM/L NaCl, 0.05% Tween 20) for 1 h at room temperature and washed in TBST. The membrane was then incubated with primary antibodies at 1:500 to 1:1000 dilutions for 1 h at room temperature or overnight at 4 $^{\circ}$ C. After washing with TBST for 15 min, the membrane was incubated with horseradish peroxidase-conjugated



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secondary antibody at a 1:5000 dilutions for 1 h at room temperature. After further washing in TBST, the proteins were detected by enhanced chemiluminescence on X-ray film with ECL Western blotting detection kit (Thermo Fisher Scientific).

2.9. Transmission electron microscopy

Treated cells were washed and fixed in 2.5% glutaraldehyde for 24 h at 4 °C and then treated with 1% osmium tetroxide, dehydrated in

Fig. 3. LFS-01 induces mitophagy in Raji and U2932 cells through activating AMPK-mTOR pathway and up-regulating p62. (a) Cells were treated with 10 μ M LFS-01 for 6 h. White circles show the colocalization of LC3 and mitochondria. Scale bar, 5 μ m. (b) Cells were treated with 10 μ M LFS-01 for 6 h. White circles show the colocalization of lysosomes and mitochondria. Scale bar, 5 μ m. (c) Colocalization coefficient between LC3 and mitochondria were computed by ImageJ program. * P < .05, ** P < .01 by paired t -test. N = 3 in triplicates. Error bars = SEM. (d) Colocalization coefficient between lysosomes and mitochondria were computed by ImageJ program. * P < .05, ** P < .01 by paired t -test. N = 3 in triplicates. Error bars = SEM. (e) Raji and U2932 cells were treated with 10 μ M LFS-01 for 0, 12, 24, and 48 h. Whole cell protein was extracted, and subjected to immunoblotting analysis. Protein level of P62, AMPK, p-AMPK (Thr172), mTOR, p-mTOR (Ser2448) were analyzed with actin loading control. (f) Quantification of the ratio of p-AMPK (Thr172)/AMPK, p-mTOR(Ser2448)/mTOR, and P62/actin, normalized for relevant control. * P < .05, ** P < .01 by paired t -test. N = 3 in triplicates. Error bars = SEM. (g) Raji and U2932 cells were incubated for 48 h with LFS-01 (10 μ M) with or without AMPK inhibitor dorsomorphin 2HCl. AMPK, p-AMPK (Thr172) and LC3-II expression were visualized by immunoblotting using specific antibodies. Actin were used as a loading control. (h) Quantification of the ratio of p-AMPK (Thr172)/AMPK, and LC3-II/actin in c, normalized for relevant control. * P < .05, ** P < .01 by paired t -test. N = 3 in triplicates. Error bars = SEM. Dors: dorsomorphin 2HCl.

a graded series of ethanol baths, infiltrated and embedded in Epon resin. Ultrathin sections of 60 nm were cut in a Leica microtome (Leica, Deerfield, III), post-stained with uranyl acetate and lead citrate, and examined in a HITACHI H-7650 transmission electron microscope (HITACHI, Ltd, Japan).

2.10. Lactate dehydrogenase release assay (LDH assay)

The release of lactate dehydrogenase was monitored using the CytoTox 96 Non-Radioactive Cytotoxicity Assay (Promega) according to the manufacturer's instructions.

2.11. RNA-seq analysis

Raji and JeKo-1 cells were exposed to vehicle (control samples) or 10 μ M LFS-01 for 48 h. Total RNA was extracted using the RNeasy kit (QIAGEN) with on-column DNase treatment. The cDNA libraries were generated using the TruSeq Sample Preparation Kit (Illumina) with polyA selection according to the manufacturer's protocol. Libraries were pooled using indexed adapters for multiplexing. Pooled libraries were normalized to 2 nM and denatured using 0.2 N NaOH prior to sequencing. Flow cell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 4000 to a depth of 200 million reads per sample. Data were analyzed using the Broad Picard Pipeline (Novogene), which includes de-multiplexing and data aggregation. The pair-ended RNA-seq raw data was examined by FastQC 0.11.5 followed by trimming to 80bp-length reads using fastx_trimmer in FastX Toolkit 0.0.13. The trimmed pair-ended RNA-seq data was mapped to pre-indexed reference human genome GRCh38plus SNPs and transcripts using HISAT2 2.0.4. The alignments were assembled, merged and quantified in StringTie 1.3.0. The expression RPM data profile of all genes in each experiment group, together with gene annotations, was extracted using SeqMonk 0.34.1. The differential expression analysis was performed using SciPy and Stats models in Python. In brief, the gene expression data of all genes in terms of RPM in distinct experiment groups were compared. The p -values were corrected using Benjamini-Hochberg in multiple tests module of Stats models to control false discovery rate. Statistical significant differential expressed genes were selected using criteria of fold-change > 2 or < 1/2, and adjusted p -value < .05. The gene ontology enrichment analysis of biological process for statistical significant differential expressed genes was performed in DAVID Bioinformatics Resources 6.8 (<https://david.ncicrf.gov/>). Among all the differential expressed genes, the autophagy-associated genes were selected and applied to Reactome FIPuIn 5.1.0. beta in Cytoscape 3.4.0 for Reactome Functional Interaction Network analysis.

2.12. Computational modeling

Crystallographic structure of CRM1 (also called as XPO1) was obtained from the PDB bank (PDB code: 3GB8). For clarity purpose, only HEAT10-HEAT13 domain of XPO1 was used in all simulation. Molecular docking was performed using our previously developed in-house docking tool FIPSDock [12]. The ligand and protein input structures in simulations were saved using PDBQT file format.

Molecular dynamics simulation was performed using academic-free package Gromacs. The molecular dynamics (MD) simulation was performed in OPLS/AA force field implemented in Gromacs. LFS-01 was parameterized following the standard protocol as described in Gromacs manual. All the simulations were performed on a high-performance DELL T7500 workstation running Linux operating system. All the molecular graphics were displayed and prepared using Pymol educational version.

2.13. Tumor xenograft study

All animal procedures in this study were approved by the Institutional Animal Care and Use Committee of local government and institutes. For mouse xenograft experiments, Raji cells (1.0×10^7) were collected and suspended in 150 μ L PBS and subcutaneously injected into the right flank of 6-week-old male BALB/c immunocompromised mice. Mice were randomized into three groups of 10 mice per group. Tumor-bearing mice were treated with vehicle or LFS-01 (twice per week, equivalent to 30 mg/kg/day) via intraperitoneal injection (i.p.). After 3 weeks, mice were sacrificed and tissues were collected for histopathological and molecular analysis. Statistically significant differences in tumor volume between the control and the treated mice were assessed by Student's t -test.

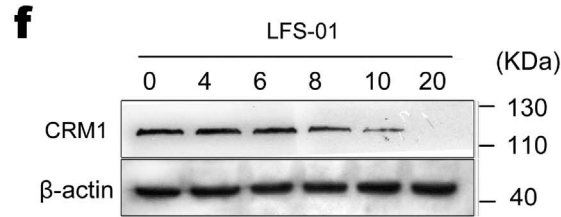
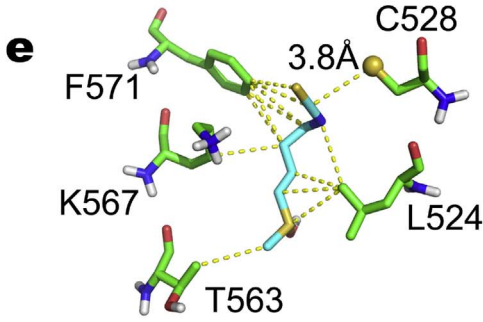
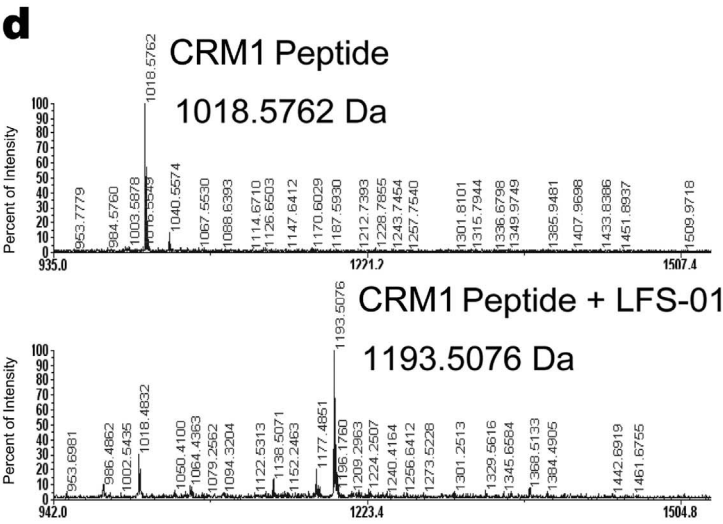
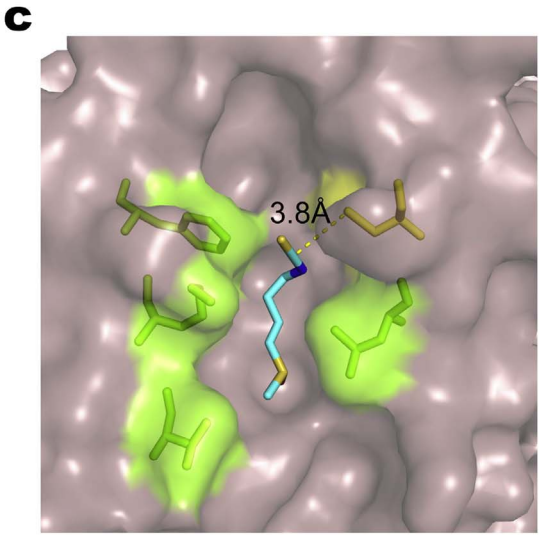
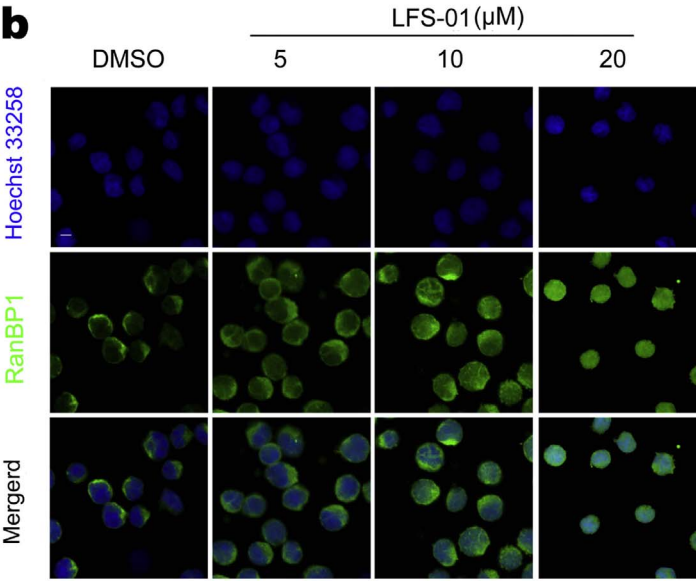
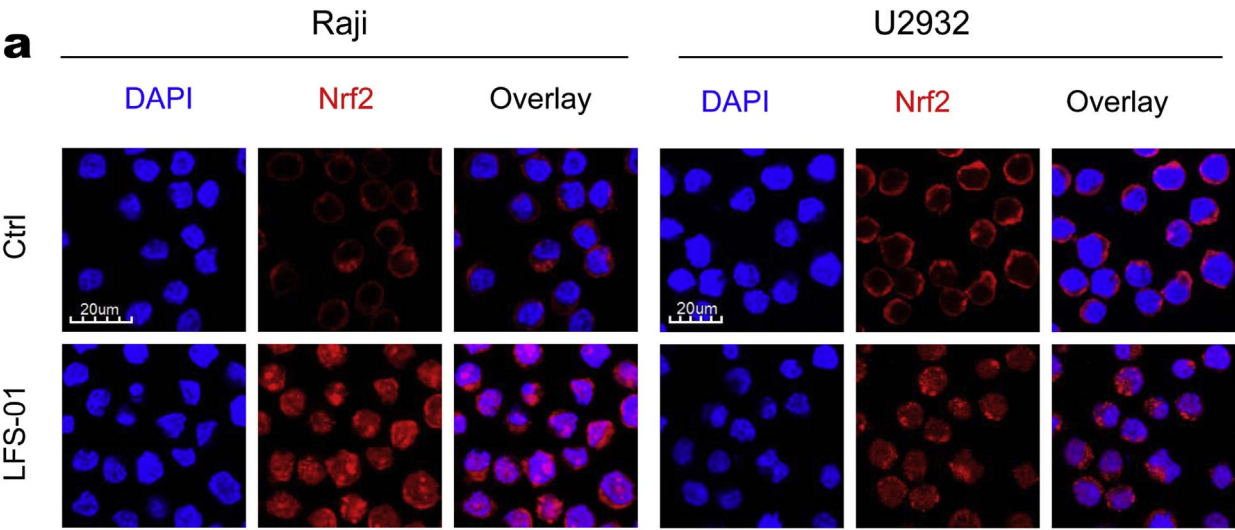
2.14. Statistical analysis

All continuous variables were compared using one-way ANOVA, followed by Dunnett's test or Tukey's test for multiple comparisons, and error bars in all figures represent the SEM. Survival curves were compared using the log-rank test with GraphPad software.

3. Results

3.1. LFS-01 induces apoptosis and cell cycle arrest in lymphoma cells

We found that LFS-01 (Fig. 1a) showed significant growth inhibition of all five lymphoma cell lines used in the experiments. The growth of five tested cell lines were less than 50% at a single dose of 5 μ M (Fig. 1b). We demonstrated that LFS-01 induced cell cycle arrest in G2/M phase in Raji and U2932 cell lines (Fig. 1c & Figure S1). And the subG1 population peak was emerged in the 20 μ M-drug-treated group, which indicated that apoptosis was induced. (Fig. S1). In fact, we showed that LFS-01 strongly induced apoptosis of lymphoma cells via flow cytometry analysis, the apoptotic cell number was increased remarkably after LFS-01 treatment (Fig. 1d & Fig. S2). The expression of p53, cleaved caspase-3, cleaved caspase-9 and cleaved PARP were elevated in a dose and time-dependent manner when treated with LFS-01 (Fig. 1e & Fig. S3). We further investigated whether Z-VAD(OMe)-FMK [13] (an anti-apoptotic peptide inhibitor of caspases) could reverse apoptosis induced by LFS-01. For this purpose, Raji and U2932 cells were pretreated with Z-VAD and then with different concentration of LFS-01. A decreased cytotoxicity against tumor cells by LFS-01 was observed yet the inhibition was only partial (Fig. S8). This suggests that LFS-01 may also exert its biological effects through other pathways.



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Fig. 4. LFS-01 up-regulates p62 through CRM1 modulation and Nrf2 nuclear retention. (a) Nuclear retention of Nrf2 after cells treated with 10 μ M LFS-01 for 24 h (b) RanBP1 nuclear aggregation in Raji cells when treated with the indicated concentrations of LFS-01 for 3 h. Nuclei were stained with Hoechst 33258. Scale bar, 5 μ m. (c) The binding configuration of LFS-01 in the groove of NES binding site of CRM1 after 10 ns MD simulation. Cys528 are depicted using a sphere model and colored yellow. (d) LFS-01 covalently binds to Cys528 of a CRM1-peptide. Synthetic peptides containing Cys528 of CRM1 or the mutant peptides in which Cys528 was substituted with Ser (supplementary Fig. S10) were treated with LFS-01 for 24 h and analyzed by MALDI-TOF MS. (e) Residues making possible contact with LFS-01 are depicted using a stick model and the interaction force between residues and LFS-01 within 4 Å are labeled in yellow dotted line. (f) Western Blot result of the protein level of CRM1 with actin as loading control. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.2. LFS-01 selectively kills lymphoma cells while sparing normal PBMC

To further assess the selectivity of LFS-01 towards lymphoma cells, we evaluated the toxicity of LFS-01 in normal PBMC isolated from peripheral blood of healthy donors (Fig. S4). Interestingly, LFS-01 barely showed any toxicity at EC₅₀ concentration of $\sim$ 5 μ M and the toxicity towards normal PBMC is minimal even at a very high dose of 40 μ M. This suggests that LFS-01 specifically eradicate lymphoma cells while sparing normal PBMC.

3.3. LFS-01 treatment strongly induces mitophagy in lymphoma cells

We wanted to determine if LFS-01 can also induce other types of cellular death in addition to apoptosis. Intriguingly, we observed massive bubble-like vesicles under microscope on the surface of U2932 (diffuse large B-cell lymphoma, DLBCL) and Raji (Burkitt's lymphoma) cells shortly after exposed to LFS-01, which indicates induction of autophagic event or necrocytosis (Fig. S5). We then wanted to examine whether the cellular death induced by LFS-01 may be caused by necrosis. We used the lactate dehydrogenase (LDH) assay [14] and the release of LDH could barely be detected after exposure to LFS-01. This result rules out the possibility that necrocytosis contributes to the death of lymphoma cells caused by LFS-01 (Fig. S6). Furthermore, we observed under transmission electron microscopy that Raji and U2932 cells exposed to LFS-01 exhibited autophagosomes and autolysosomes containing digested mitochondria and cytosolic fragments (Fig. 2a–d). More interestingly, we observed extensive fragmentation of mitochondria after the addition of LFS-01 into these two cell lines. We found that Raji and U2932 cells treated with LFS-01 showed a time-dependent elevation of the biomarkers for autophagy [15] (namely the ratio of LC3A-II/ β -actin in the total cell lysate) (Fig. 2e). The induction of autophagy by LFS-01 was also revealed by monitoring the accumulation of punctate LC3 using confocal microscopy (Fig. 2f). We further ascertained the presence of mitophagy induced by LFS-01 using two complementary methods. First, we observed a colocalization of LC3 with a fragmentation of the mitochondria network after LFS-01 treatment visualized by confocal microscopy with mitotracker dye (Fig. 3a and c). This is a clear indication that mitophagy may have occurred [16]. Second, we found a similar colocalization of lysosomes with a fragmentation of the mitochondria network after LFS-01 treatment visualized by confocal microscopy with mitotracker dye and lysotracker dye (Fig. 3b and d). This is in accordance with the mitophagy process during which the autophagosomes containing mitochondria fragmentation fused with lysosomes to form autolysosomes for further degradation [17]. These two complementary experiments strongly indicated that LFS-01 induced mitophagy in both Raji and U2932 cells.

Moreover, we want to remind the reader that we also observed the formation of double- or multiple-membraned vesicles after LFS-01 treatment as evident by the TEM image, a typical phenotype of macroautophagy. Therefore, this implicates that LFS-01 may also induce macroautophagy in the cells and in animals. Further experiments are warranted to confirm our observation.

3.4. LFS-01 exerts therapeutic effects by triggering concomitant mitophagy and apoptosis

We wanted to investigate whether the suppression of both mitophagy and apoptosis could change the ratio of viable cells that had been

treated with LFS-01. For this purpose, both Z-VAD (a caspase inhibitor) and 3-methyladenine (3-MA) [20] (an autophagy inhibitor) were used in the experiments. First, LFS-01 treatment of lymphoma cells (Raji and U2932) pretreated with 3-MA showed a significant decrease in cytotoxicity against tumor cells. Furthermore, as expected, the pretreatment with both 3-MA and Z-VAD simultaneously nearly blocked the cytotoxicity of LFS-01 against lymphoma cells (Fig. S4). These results strongly suggest that LFS-01 exerts its therapeutic effects by concomitant mitophagy and apoptosis.

3.5. LFS-01 induces mitophagy through the CRM1-mediated p62 overexpression and activation of AMPK/mTOR pathway

Of note, p62 is a central player in the mitophagy process for the elongation and accumulation of autophagosomes [21]. Prior studies have shown that the accumulation of p62 could promote mitophagic clearance of damaged mitochondria [4]. In consistent with previous reports, we observed that LFS-01 can upregulate p62 in lymphoma cells in time-dependent manner (Fig. 3e–f). On the other hand, recently, numerous studies suggest that AMPK activation could induce mitochondrial fragmentation and lead to mitophagy through inhibition of mTOR as well as regulation of a few other governing factors during the mitophagic process [22]. Therefore, we want to examine if the induction of mitophagy by LFS-01 was also caused by the engagement of AMPK/mTOR pathway. First, we revealed that the ratio of expression level of p-AMPK/AMPK was elevated in time-dependent manner after treatment of LFS-01 compared to control cells (Fig. 3e–f). Subsequently, we demonstrated by western blot that LFS-01 treatment decreases the ratio of p-mTOR/mTOR (Fig. 3e–f), indicating that LFS-01 may trigger mitophagy through activating AMPK and inhibiting mTOR. Furthermore, we showed that the suppression of AMPK by its inhibitor dorsomorphin in both lymphoma cell lines led to a significant abrogation of LC3-II accumulation, indicating the crucial role of AMPK in LFS-01-mediated mitophagy (Fig. 3g–h). Importantly, we revealed by confocal microscopy that LFS-01 treatment can cause nuclear retention of Nrf2 (Fig. 4a). CRM1 is a protein (also called as XPO1) responsible for the nuclear traffic of numerous tumor suppressor proteins and transcription factors such as p53, Foxo1 and Nrf2 from nucleus to cytoplasmic compartment [23]. Therefore, we concluded that the overexpression of p62 after LFS-01 treatment may be due to the inhibition of CRM1 and subsequent nuclear accumulation of Nrf2, a transcription factor that regulates the expression of a range of ARE-dependent gene products including p62 [24]. Collectively, these results suggested that the accumulation of p62 and the engagement of AMPK/mTOR pathway are both responsible for the induction of mitophagy by LFS-01 treatment.

3.6. LFS-01 modulates CRM1 by covalently binding to a conserved cysteine residue

We want to examine if LFS-01 can directly target at CRM1 (or XPO1) and thereby regulate nuclear export. First, we revealed that RanBP1, a Ran-binding protein consisting of NES and a canonical biomarker for CRM1 inhibition, was retained in the nucleus in a dose-dependent manner upon treatment of LFS-01 (Fig. 4b). Moreover, we showed that LFS-01 treatment causes nuclear accumulation of p53 in Raji and U2932 cell line (Fig. S9). Noteworthy, it has been well established that the nuclear accumulation of p53 could lead to apoptosis in tumor cells. Second, we demonstrated that LFS-01 treatment

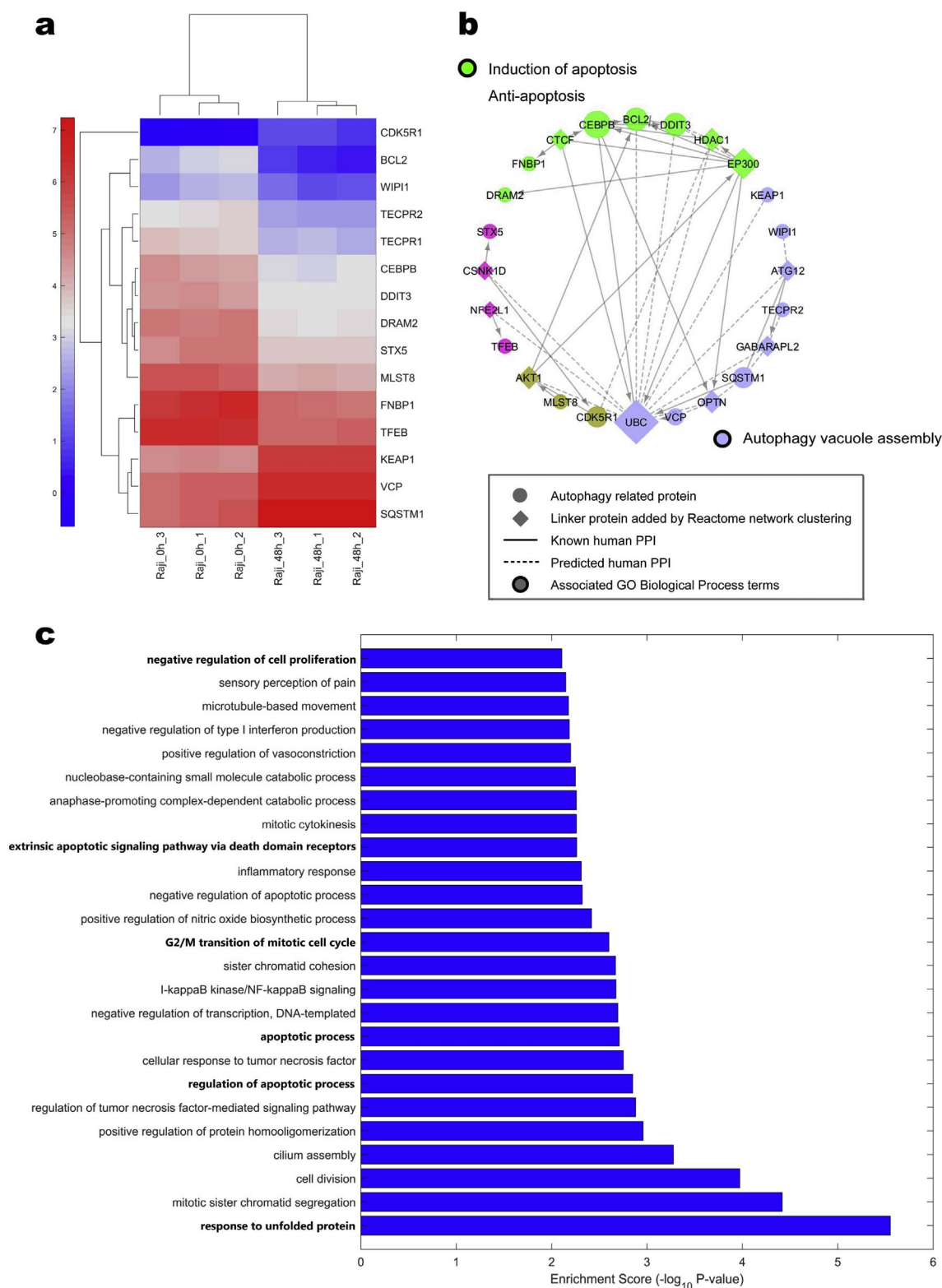


Fig. 5. Differential gene expression analysis of LFS-01-treated Raji cells. (a) Heatmap for 15 differentially expressed autophagy-associated genes in Raji cells at 0 h and 48 h. Unsupervised clustering on 15 differentially expressed genes (log₂ transformed RPKM) grouped LFS-01 treated and normal samples together. The 15 genes were *CDK5R1*, *BCL2*, *WIP1*, *TECPR2*, *TECPR1*, *CEBPB*, *DDIT3*, *DRAM2*, *STX5*, *MLST8*, *FBNP1*, *TFEB*, *KEAP1*, *VCP* and *SQSTM1*. (b) Network representation of 15 of 211 autophagy-associated proteins, clustered into functional modules of human protein-protein interactions by the Reactome Functional Interaction Network. Edges represent human protein-protein interactions (PPI) reported in the Reactome database. Cellular functional compartments that are important in autophagy modulation are indicated by color-coded modules annotated by module-enriched GO Biological Process terms (FDR < 0.01). Node size ranks proteins by PPI count. (c) GO enrichment analysis of genes regulated by LFS-01 in Raji cell at 48 h. The GO terms are sorted by -log₁₀ of the enrichment P-value, which represents the enrichment significance of GO terms. The enrichment of GO terms is shown by comparing all DEG with the whole genome. Pathways related to cell proliferation, cell cycle, cell apoptosis and cell autophagy are highlighted in bold. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

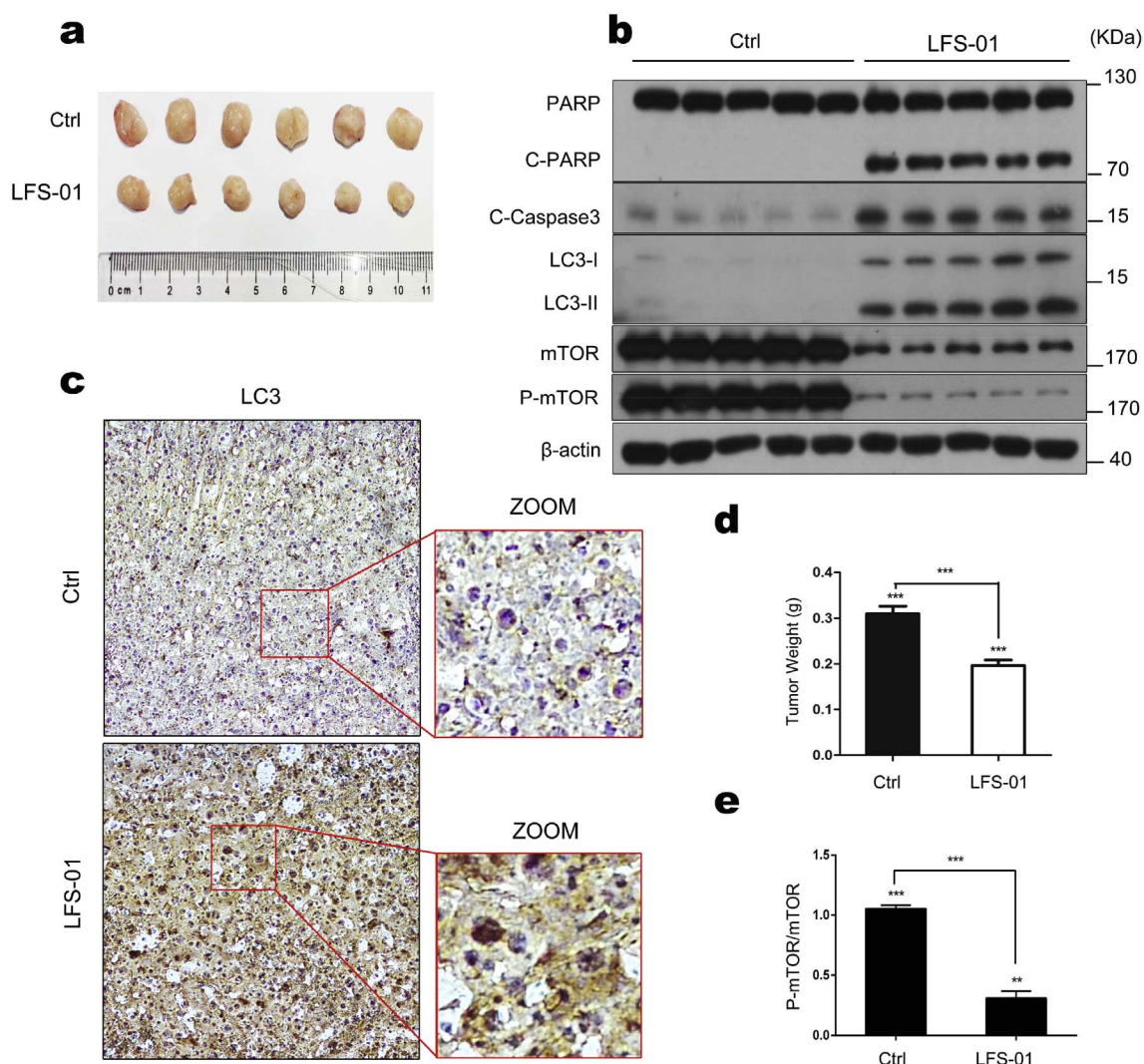


Fig. 6. Suppression of lymphoma cell proliferation and xenograft tumorigenesis by LFS-01. (a) Raji cells were subcutaneously injected into BALB/c male nude mice. After two weeks, mice were sacrificed and tumors were collected to assay the tumor weight. (b) Whole cell proteins were extracted from the xenograft tumors, and subjected to Western blot of PARP, C-Caspase3 (cleaved caspase3), LC3, mTOR and p-mTOR (Ser2448). Actin serves as loading control. (c) Immunohistochemical staining of LC3 in tumors of control and LFS-01 treated mice. (d) Significant differences of tumor weight between LFS-01 pretreated and control samples. * $P < .05$, ** $P < .01$, *** $P < .001$ by one-way ANOVA, followed by Dunnett's test or Tukey's test. $N = 5$. Error bars = SEM. (e) Quantification of the ratio of p-mTOR (Ser2448)/mTOR, normalized for relevant control. * $P < .05$, ** $P < .01$ by paired t -test. $N = 5$. Error bars = SEM.

diminished the expression of CRM1/XPO1 in Raji cells in a dose-dependent manner (Fig. 4f). Furthermore, we want to examine if LFS-01 could bind covalently to the nucleophilic sulfhydryl (-SH) group of a conserved cysteine⁵²⁸ residue in the NES-binding groove of CRM1 protein by matrix-assisted laser desorption ionization-time of flight mass spectrometry (MALDI-TOF) technique. Similar to previous studies [25,26], we synthesized both the wild type (WT) peptide containing Cys and the mutant peptide (MT) containing Ser at the position corresponding to Cys⁵²⁸ of CRM1. Mass spectrometric analysis of the wild type peptide revealed a mass peak at m/z 1018.5762. Interestingly, upon the incubation with LFS-01, additional peak at m/z 1193.5076 (+175, the molecular weight of LFS-01) was detected which corresponds to the covalently adduct of the peptide with LFS-01 (Fig. 4d). On the other hand, no covalently adduct was observed for the sample derived from the MT peptide incubated with LFS-01 (Fig. S10). Finally, we prepared Raji cells transfected with a CRM1 mutant expressing a Cys528Ser substitution. This is because cells expressing the mutant as mentioned above are viable yet insensitive to the treatment of nuclear export modulators. Intriguingly, we found that cells expressing HA-CRM1-C528S mutant were rather resistant to the treatment of LFS-01 (Fig. S11). In addition, we employed computational study to rationalize

the possible binding mode of LFS-01 at the NES-binding cleft of CRM1. Computational analysis showed that LFS-01 can stably fit into the hydrophobic groove of NES-binding cleft of CRM1 as judged by shape complementarity and putative interactions (Fig. 4c). In particular, the Michael acceptor moiety of LFS-01, the isothiocyanate group is within the 'reactive distance' of the sulfur atom of Cysteine residue to enable Michael reaction (Fig. 4e). These results strongly suggest that LFS-01 can modulate nuclear export by directly targeting the sulfhydryl (-SH) group of a conserved cysteine residue in the NES-binding domain of CRM1.

3.7. LFS-01 exposure modulates the mitophagic signaling at transcriptional level

To further delineate the mechanism accounting for the mitophagic event induced by LFS-01, we performed RNA-seq experiments and bioinformatics analysis for Raji cell line (Fig. S12). In brief, we performed a differential abundance analysis between untreated Raji cells and Raji cells exposed to LFS-01 (10 μ M). In total, we identified 1391 genes differential expressed at transcript level. Noteworthy, a recent study reported the curation of 211 autophagy-associated (AA) human

genes from the literature [27]. We then compared the 1391 differentially expressed genes with the 211 autophagy-associated genes and identified 15 overlapped AA genes including: *CDK5R1*, *BCL2*, *WIP1*, *TECPR2*, *TECPR1*, *CEBPB*, *DOIT3*, *DRAM2*, *STX5*, *MLST8*, *FBNP1*, *TFEB*, *KEAP1*, *VCP* and *p62/SQSTM1* (Fig. 5a). In particular, among the 15 autophagy-associated genes we found *p62/SQSTM1*, a selective autophagy receptor which was upregulated upon LFS-01 treatment. This was corresponding with our experimental result mentioned before that LFS-01 could upregulate p62 protein level (Fig. 3e). Indeed, *p62/SQSTM1* is a crucial mediator for selective autophagic process including mitophagy. Namely, the binding of *p62/SQSTM1* to ubiquitinated mitochondrial proteins is crucial to tether the damaged mitochondria to the LC3-positive phagophore for engulfment during mitophagy process [4,28]. Our results confirmed the hypothesis that *p62/SQSTM1* is indispensable for mitophagy event. In addition, we found that valosin-containing protein (*VCP*, also known as p97), a class II member of the ATPase responsible for mitochondria-associated degradation as well as mitophagy [29,30], is significantly upregulated after LFS-01 treatment. This is consistent with our findings of excessive mitochondrial fragmentation upon LFS-01 exposure. Moreover, *BCL2*, a well-known anti-apoptotic protein as well as an autophagy inhibitor [31], was significantly downregulated after LFS-01 treatment. Our protein interactome network analysis (Fig. 5b) and gene ontology (GO) enrichment analysis (Fig. 5c) (The GO terms related to cell proliferation, cell cycle, cell apoptosis and cell autophagy are highlighted in bold.) further revealed that the events of apoptosis and the assembly of autophagy vacuole were significant upon LFS-01 exposure.

3.8. LFS-01 administration significantly suppresses tumor growth in rodent model

Next, we wanted to assess LFS-01 as a single therapeutic agent in the nude mouse xenograft model bearing Burkitt's lymphoma. Treatment with LFS-01 (i.p. administration, dose: 100 mg/kg twice a week, equivalent to 30 mg/kg/day) resulted in a dramatic suppression of tumor growth after 10 days of drug exposure compared to vehicle-treated group (Fig. 6a and d). These events became more apparent following 20 days of LFS-01 treatment. However, no significant change of body weight was noted compared to vehicle control group, suggesting a low toxicity profile of LFS-01. Furthermore, we determined both mitophagy and apoptosis in tumor tissue of xenograft tumors. As expected, immunohistochemical staining analysis revealed that LC3 expression was significantly increased in xenograft tumor sections from mice treated with LFS-01 in comparison to the vehicle-treated group. In addition, Western blot analysis also revealed that cleaved PARP and caspase 3 were elevated in xenograft tumor tissue after LFS-01 treatment compared to control group (Fig. 6b). On the other hand, both ratios of LC3-I/LC3-II and p-mTOR/mTOR were significantly decreased upon LFS-01 treatment compared to control group (Fig. 6b and e). These findings strongly implicated that concomitant mitophagy and apoptosis are critical mechanisms accounting for the selective eradication of lymphoma cells by LFS-01.

4. Discussion

Although enormous efforts have been made towards developing immunotherapies for liquid tumors, lymphoma especially non-Hodgkin lymphomas (NHL) still represent as a serious threat to human health. Particularly, patients with aggressive forms of NHL such as diffuse large B cell lymphoma (DLBCL) and Burkitt's lymphoma usually have a poor prognosis with limited treatment options [32]. In this respect, for the first time, we report that traditional Chinese medicine-derived LFS-01 exerts its biological effects by triggering a concomitant mitophagy and apoptosis in lymphoma cells. We showed that LFS-01 displayed strong in vitro and in vivo anti-neoplastic properties against NHL with minimal toxicity in normal cells, suggesting possible weak adverse

effects of this compound.

Thus far, three types of programmed cellular death have been described, namely apoptosis, autophagy and necrosis. Herein, the relationship between apoptosis and autophagy is rather complex and largely dependent on cell type and stimuli. Nevertheless, mounting evidences suggest that apoptosis (self-killing) and autophagy (self-eating) are mechanistically linked. Interestingly, it has been suggested that therapeutic strategies that can trigger both apoptosis and autophagy in tumor cells may be advantageous such that non-specific toxicity can be avoided [33]. Moreover, synergistic effects of autophagy and apoptosis may contribute to the selective killing of tumor cells [34,35]. To date, several naturally-derived compounds have been reported to induce both autophagy and apoptosis in tumor cells such as arsenic trioxide [36], vitamin K2 [37], and resveratrol [38]. Notably, arsenic trioxide is also derived from traditional Chinese medicine and has been used in clinical practice for treating acute myeloid leukemia (AML) [39]. In this study, we found that LFS-01 treatment of NHL cell lines (Raji and U2932) pretreated with either z-VAD (a caspase inhibitor) or 3-MA (an autophagy inhibitor) resulted in a significant reduction of cytotoxicity against tumor cells. Further, we observed that the killing of tumor cells by LFS-01 was nearly blocked when the lymphoma cells were simultaneously pretreated with both z-VAD and 3-MA. This strongly suggests that LFS-01 kills lymphoma cells through the activation of both mitophagy and apoptosis. Interestingly, we examined that LFS-01 has minimal cytotoxicity against normal lymphocytes from healthy donors, clearly demonstrating its selective eradication of lymphoma cells. We subsequently assessed the in vivo antitumor activities of LFS-01 in xenograft model. LFS-01 was able to significantly reduce the tumor volume in SCID mice that received a subcutaneous injection of lymphoma cells. Moreover, histopathological analysis revealed that cleaved PARP and caspase 3 were elevated after LFS-01 treatment. On the other hand, both ratios of LC3-I/LC3-II and p-mTOR/mTOR were significantly decreased upon LFS-01 administration, supporting the in vitro results that LFS-01 exerts therapeutic effects through triggering dual mitophagy and apoptosis.

Mitophagy is a selective autophagic process which leads to the degradation of damaged mitochondria in response to metabolic or environmental stress conditions. In this study, we found that LFS-01 treatment significantly enhances p62 in a time-dependent manner. We found that the upregulation of p62 is due to the modulation of CRM1 and subsequent nuclear accumulation of cargo protein Nrf2, a transcription factor that triggers and regulates p62 expression [40,41]. Our results are in line with prior reports that p62, an autophagic adaptor, is indispensable for the clearance of defective mitochondria during mitophagic events [42]. Namely, the accumulation of p62 is closely associated with the induction of mitophagic signaling. Noteworthy, our study is the first to establish the functional link between CRM1-mediated upregulation of p62 and mitophagic cellular death in tumor cells. In addition, we found that nuclear p53 of lymphoma cells were increased in dose-dependent manner after LFS-01 treatment. Despite the well-documented role of p53 as a tumor suppressor, accumulating evidence suggest that p53 present in cytoplasmic compartment can suppress mitophagy. Therefore, we want to argue that nuclear accumulation of p53 caused by LFS-01 treatment may counteract the suppressing effects of cytoplasmic p53 towards mitophagy. More importantly, we showed that AMPK, a primary inducer of mitophagy in response to mitochondria stress, was activated upon LFS-01 exposure. This is in accordance with recent findings that the activation of AMPK is directly associated with mitochondria fission followed by mitophagy events [8]. Furthermore, we demonstrated that mTOR, a central signal integrator and a negative regulator of mitophagy, was inactivated upon LFS-01 treatment. This is consistent with previous research that activation of AMPK can lead to inhibition of mTOR and induction of mitophagic signaling. Collectively, our results demonstrated that concomitant mitophagy and apoptosis are crucial mechanisms accounting for the selective killing of lymphoma cells by LFS-01 through the CRM1-

mediated overexpression of p62 and engagement of AMPK/mTOR pathway.

The functional crosstalk between autophagy (including mitophagy) and apoptosis remains as a topic of much debate. In some scenarios, autophagy or mitophagy plays a protective role for cells in response to a stress or stimuli whereas in other cellular settings, autophagy or mitophagy constitutes an alternative or synergistic pathway leading to cellular demise [43]. The induction of autophagic or mitophagic cell death could be particularly useful and may constitute a ‘fail safe’ mechanism when the cellular death by apoptosis is inefficient or defective. Hence, therapeutics that can trigger both mitophagy and apoptosis may represent a very attractive approach for selective and efficient killing of tumor cells, as several studies have suggested [44,45]. To this end, our study reported a naturally-derived compound that enables a functional cooperation and synergistic effects between mitophagy and apoptosis for the efficacious killing of liquid tumor cells.

5. Conclusions

In summary, we present herein naturally-derived compound LFS-01 can induce both mitophagy and apoptosis which accounts for the selective eradication of lymphoma cells. The mechanisms involve the CRM1-mediated overexpression of p62 and activation of AMPK/mTOR pathway. Our results link the mechanisms of anticancer effects of LFS-01 with the induction of mitophagy and highlight the significance of mitophagy in selective killing of cancer cells. These findings provide a strong rationale to developing innovative therapies for cancer prevention and treatment based on dual mitophagy and apoptosis.

Compliance with ethical standards

Disclosure of potential conflicts of interest: Lei Mao serves as Chief Computing Officer for DrivingForce Therapeutics (DFT); Yongliang Yang is a scientific co-Founder of DFT and Cheng Li is a scientific advisor to DFT.

Research involving human participants and/or animals: All applicable international, national, and/or institutional guidelines for the care and use of animals were followed; All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent: Informed consent was obtained from all individual participants included in the study.

Author contributions

H.W. and Y.Y. designed research and performed data analysis for the entire project. H.W. led the project and prepared the figures and tables for this manuscript. F.W. performed animal experiments and J.Z. participated in data analysis. S.W. designed the study of molecular dynamics simulation and performed data analysis. Z.L., T.L. and C.L. performed next-generation sequencing and data analysis. C.G.L. helped in the study of lymphocytes isolated from peripheral blood of healthy donors. L.M. participated in the design and conception of the whole project, performed bioinformatics and statistical analysis for the manuscript. Y.Y. oversaw the whole project and wrote the manuscript.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.cbi.2017.12.017>.

Transparency document

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